Assignment 4 – Profit & Loss

Q1) An article when sold for Re 4600 makes a 15% profit. Find the profit or loss percent. If it was sold for Re 3600.

→ 15% profit on 4600

Cost Price = $\frac{\text{Selling Price} \times 100}{100 + \text{gain}\%}$

↳ $= \frac{4600 \times 100}{100+15} = \frac{4600 \times 100}{115} = \underline{\underline{4000}}$ (C.P)

∴ Loss of Rs 400/- when sold for Rs 3600

∴ % of loss $= \frac{400 \times 100}{4000} = \underline{\underline{10\%}}$ Loss

Q2) When a man sold an article for Rs 540, he made a loss of 10%. At what price should he sell it, so he incurs a loss of only 5%.

→ Here, Selling Price = Rs. 540

Loss % = 10%

∴ Cost Price (C.P) $= S.P \times \frac{100}{(100-10)} = 540 \times \frac{10\cancel{0}}{9\cancel{0}}$

$= \frac{5400}{9} = \underline{\underline{600}}$ (C.P)

∴ S.P of the article at 5% loss : $C.P \times \frac{(100-5)}{100}$

$= \frac{\overset{6}{\cancel{600}} \times 95}{\cancel{100}} = \underline{\underline{570}}$ (5% loss)

Q3) Ram sells chocolates at a profit of 20% for Rs 60. What will be the percentage loss or gain, if he reduces price to Rs 55 due to less demand?

→ Original C.P $= S.P \times \frac{100}{(100+20)} = \frac{\cancel{60} \times 100}{\cancel{120}\,2} = \underline{\underline{50}}$ (C.P)

∴ It will be profit percentage if sold for Rs 55.

∴ The profit percentage $= \left(\frac{\text{Gain} \times 100}{C.P}\right)$

$= \frac{5 \times \cancel{100}^{2}}{\cancel{50}_{1}} = \underline{\underline{10}}$ (Profit percent)

Q4) A man sold two plots for Rs 8 lakhs each. On one he earns a profit of 16% and on the other he loses 16%. How much loss did he loss or gain whole transaction?

→ Apply formula, (At this scene, there is always loss)

$$\text{Loss\%} = \left(\frac{\text{Common loss \& gain\%}}{10}\right)^2 \%$$

$$\rightarrow = \left(\frac{\cancel{16}^{8}}{\cancel{10}_{5}}\right)^2 \% = \left(\frac{64}{25}\right)\% = \underline{2.56\%}$$

Q5) A shopkeeper sells an article for Rs 200 with a loss of 20%. Find the cost price of the article.

→ S.P = 200 Loss = 20%

C.P = ?

$$\therefore \text{C.P} = \frac{100}{(100-20)} \times 200$$

$$= \frac{100}{\cancel{80}_{2}} \times \cancel{200}^{5} = 50 \times 5 = \underline{250} \text{ (C.P)}$$

Q6) A trader expects a gain of 15% on his cost price. If in a week his sale is of Rs 580, then what is his profit?

→ S.P = 580 Expected profit = 15%

C.P = ?

$$\therefore \text{S.P} = \text{C.P} + \text{Profit}$$

$$= \text{C.P} + \left(\frac{15}{100} \times \text{CP}\right)$$

$$580 = \text{C.P} + (0.15\ \text{C.P})$$

$$580 = 1.15\ \text{C.P}$$

$$\therefore \text{C.P} = \frac{580}{1.15} = \underline{504.35} \text{ (C.P)}$$

Now, Profit = S.P − C.P

= 580 − 504.35

= <u>75.65</u> (Profit)

Q7) If a boy sells a book for Rs 450, he gets a loss of 10%, then find the Cost Price. To ~~10%~~ gain 10%, what should be selling Price?

→ S.P = 450 → Loss = 10%

C.P = ?

$\therefore$ C.P = $\frac{100}{(100-10)} \times 450$

$= \frac{100}{90} \times 450 = 500$ (C.P)

Now,

C.P = 500 gain % = 10%

S.P = ?

$\therefore$ S.P = $\frac{(100+10)}{100} \times 500$

$= \frac{110}{100} \times 500 = 550$ (10% profit)

Q8) S.P of 10 candles is same as C.P of 12 candles. Find the gain percent.

→ S.P of 10 candles = C.P of 12 candles

Gain % = ?

Let the S.P of one candle be x

$\therefore$ S.P of 10 candles = $10x$

$\therefore$ C.P of 12 Candles = $10x$

C.P of 1 candle = $\frac{10x}{12} = 0.8x$

Profit % = $\frac{(x - 0.8x) \times 100}{x}$

$= \frac{0.2x \times 100}{x}$

$= 20\%$

(No. of Y articles > No. of X articles)

Profit % = $\frac{\text{No. of X articles} - \text{No. of Y articles}}{\text{No. of Y articles}} \times 100$

$= \frac{12 - 10}{10} \times 100$

$= \frac{200}{10} = 20\%$

Q9) The selling price of 40 apples is equal to the 35 apples. Find the profit or loss % obtained?

→ Let the C.P of each apple be Rs 1.

Given, S.P of 40 apples = C.P of 35 apples

∴ S.P of 40 apples = 35

Loss = C.P − S.P = 40 − 35 = 5

$\therefore$ Loss % $= \frac{\cancel{5}^{1} \times 100}{\cancel{40}_{8}} = \underline{\underline{12.5\%}}$

Q10) A man purchased two plots Rs 5,00,000. On one he gains 15% while on other he ~~at~~ loses 15%. Find how much does he gain or lose during this transaction?

→ In such cases, mostly there is loss.

$\therefore$ Loss % $= \left(\frac{\text{Common loss \& gain \%}}{10}\right)^2$

$\hookrightarrow = \left(\frac{\cancel{15}^{3}}{\cancel{10}_{2}}\right)^2 = \left(\frac{3}{2}\right)^2 = \frac{9}{4} = \underline{\underline{2.25\%}}$

Q11) A boy bought a horse and carriage for Rs 5000. He sells the horse at a gain of 20% and the carriage at the loss of 10%. If he gains 3% on whole, find the cost price of the horse.

→ Let the price of horse be x

Cost of horse & carriage = Rs. 5000

$\therefore$ Cost of carriage = $(5000 - x)$

After selling horse → Gains 20%

After selling carriage → loss 10%

Overall gain → 3% gain

$\therefore$ (20% of x) − (10% of carriage) = 3% gain

$= \frac{20}{100} * x - \frac{10}{100} * (5000 - x) = \frac{3}{100} * 5000 = 50$

$= \frac{x}{5} - \frac{(5000 - x) * 10}{10} = 150$

Multiply both sides by 10, for substraction:

$$\frac{10x}{5} - \frac{(5000 - x) * 10}{10} = 150 * 10$$

$$2x - 5000 + x = 1500$$

$$3x - 5000 = 1500$$

$$3x = 1500 + 5000$$

$$3x = 6500 \quad \therefore x = 2166.67$$

(Q12) A dishonest shopkeeper sells his grocery using weights 10% less than true weights and makes a profit of 30%. Find his total gain percentage.

→ Assuming, the grocery bag be 1000 gm (True C.P)

∴ By using false weights, 10% less

∴ 90% of 1000 gm = 900 gm (C.P)

Considering 1 gm = Rs. 1, 900 gm = Rs. 900

He sells with gain of 30% on true C.P

$$\therefore S.P = \frac{(100 + \text{Gain}\%)}{100} \times \underset{\text{(True C.P)}}{C.P}$$

$$= \frac{(100+30)}{100} \times 1000 = 1300$$

$$\therefore \text{Gain} = S.P - C.P$$

$$= 1300 - 900$$

$$= 400$$

$$\therefore \text{Gain}\% = \frac{400}{900} \times 100 = 44.44\%$$

Q13) The owner of a furniture shop charges his customer 24% more than the Cost Price. If a customer paid ₹8339 for a computer table, then what was the price of computer table?

→ S.P = 8339 → Profit % = 24%

C.P = ?

$\therefore CP = SP * \frac{100}{(100+24)}$

$= 8339 \times \frac{100}{124} = \frac{833900}{124}$

$= \underline{6725}$ (C.P)

Q14) Sreenivas sells a table to Shiva at 10% profit and Shiva sells it to Mahesh at 10% loss. At what price did Sreenivas purchased the table if Mahesh paid Rs. 2178?

→ Let the C.P of table for Sreenivas be x.

C.P of table for Mahesh = 2178

Sreenivas sold table at 10% profit to Shiva,

Shiva sold table at 10% loss to Mahesh, Mahesh paid 2178.

$\therefore$ (90%) of (110%) of x = 2178

$= \left(\frac{90}{100}\right)\left(\frac{110}{100}\right)x = 2178$

$= \left(\frac{110}{100}\right)x = 2178\left(\frac{10}{9}\right)$

$= \left(\frac{11}{10}\right)x = 2178\left(\frac{10}{9}\right)$

$= x =$

$= \left(\frac{90}{100}\right)\left(\frac{110}{100}\right)x = 2178$

$= x = \frac{2178 \times 100}{9 \times 11}$

$x = 2200$

Q 15) The value of machine depricates at 20% per annum. If present value is 1,50,000 Rs, then at what price it should be sold after two years such that profit of 24000 is made?

→ Current value of machine = 1,50,000 Rs

Thus value of machine after 2 years

$$= \frac{80}{100} * \frac{80}{100} * 1,50,000$$

$$= 96,000$$

∴ S.P such that profit of 24,000 is made,

$$96000 + 24000 = 1,20,000$$

Q16) The sale price of an article including the sale tax is Rs. 616. The rate of sales tax is 10%. If the shopkeeper has made the profit of 12%, then what is C.P of article?

→ Let the C.P be x

∴ S.P = x + 12% of x

$$S.P = \frac{(100+12)}{100} x$$

$$S.P = \frac{112}{100} x = \frac{28x}{25} \quad (S.P)$$

S.P of article including sale tax is Rs. 616.

Sale tax is 10% of S.P

∴ S.P + 10% of S.P = 616

$$\frac{28x}{25} + 10\% \text{ of } \frac{28x}{25} = 616$$

$$\frac{28x}{25} + \left(\frac{\cancel{10}^{1}}{\cancel{100}_{10}}\right)\left(\frac{28x}{25}\right) = 616$$

$$\frac{28x}{25} + \frac{28x}{250} = 616$$

$$\frac{280x}{250} + \frac{28x}{250} = 616 = \frac{308x}{250} = 616$$

$$= x = \frac{\cancel{616}^{2} \times 250}{\cancel{308}_{1}} = x = 500 \quad (C.P)$$

(Q17) Jacob bought a scooter for certain sum of money. He spent 10% of the cost on repairs and sold the scooter for a profit of Rs. 1100. How much did he spend on repairs if he made a profit of 20%?

→ Let the C.P of scooter be x,

Jacob made profit of 20% on 1100 Rs

$$\therefore \frac{20}{100} x = 1100$$

$$x = \frac{1100 \times 100}{20} = \underline{\underline{5500}} \text{ (C.P)}$$

+ repairs

$\therefore$ 10% of C.P spent on repairs

$$\therefore \text{Actual price} = \left(\frac{100}{100+10}\right) * \text{C.P}$$

$$= \left(\frac{100}{110}\right) * 5500 = \underline{\underline{5000}}$$ (Actual C.P)

$$\therefore \text{Spent on repairs} = 5500 - 5000$$

$$= \underline{\underline{500}} \text{ R.S}$$

(Q18) The ratio of cost price and selling price is 4:5. The profit percent is:

→ C.P and S.P ratio = 4:5

Let C.P = $4x$ and S.P = $5x$

$$\therefore \text{Profit} = \text{S.P} - \text{C.P}$$

$$= 5x - 4x$$

$$= x$$

$$\text{Profit \%} = \left(\frac{\text{Profit} \times 100}{\text{C.P}}\right)$$

$$= \frac{x \times 100}{4x} = \frac{100x}{4x} = \underline{\underline{25\%}}$$

Q19) In a certain store, the profit is 320% of the cost. If the cost increases by 25% but the selling price remains constant, approximately what percentage of selling price is profit?

→ Let the C.P be ₹100.

The Profit = Rs320

$\therefore$ S.P = 100 + 320

= 420

New C.P increases by 25%

$\therefore$ New C.P = Rs 125

But S.P is constant, SP = 420

$\therefore$ Profit = S.P − C.P

= 420 − 125

= 295

$\therefore$ Required percentage $= \frac{295}{420} \times 100$

= Approx. 70%

Q20) The profit earned by selling an article for Rs. 832 is equal to the loss incurred when the same article is sold for Rs. 448. What should be the sale for making 50% profit?

→ Let profit and loss be x,

Then, $832 - x = 448 + x$

$-x - x = 448 - 832$

$-2x = -384$

$\therefore x = \frac{-384}{-2} = 192$ Rs

$\therefore$ C.P of article = 832 − 192 = 640 Rs (C.P)

$\therefore$ S.P at 50% profit $= C.P * \left(\frac{100+50}{100}\right)$

$= 640 * \left(\frac{\cancel{150}^{3}}{\cancel{100}_{2}}\right) = \frac{\cancel{640}^{320} \times 3}{2} = 960$ Rs

(S.P at 50% profit)

Q21) The cash difference between the selling price of an article at a profit of 4% and 6% is Rs. 3. The ratio between two selling prices is:

→ Let the C.P of artice be Rs. 100

∴ S.P of first article $= 100 \times \left(\frac{100+4}{100}\right) = \underline{\underline{104}}$

S.P of second article $= 100 \times \left(\frac{100+6}{100}\right) = \underline{\underline{106}}$

∴ The ratio of S.P of first article to second article

$= 104 : 106$

$= \underline{\underline{52 : 53}}$